

RASHID Supply Chain Copilot

Executive Product Brief

Product Name: RASHID (رشيد) — Arabic for "The Wise/Prudent" **Organization:** Al Ghurair Investment (AGI) — Food Division **Division Brands:** Grand Mills, Jenan, Animal Feed, Specialty & Industrial **Version:** 1.0 Prototype **Date:** May 2026

1. Executive Summary

RASHID is an AI-powered supply chain copilot purpose-built for AGI Food Division — the UAE's leading flour miller and branded food manufacturer operating Grand Mills (800+ MT/day milling capacity across Dubai and Abu Dhabi), Jenan consumer brands (flour, pasta, oils, rice, sugar), Animal Feed, and Specialty/Industrial ingredients.

RASHID unifies demand forecasting, inventory management, supplier monitoring, production scheduling, and scenario planning through a multi-agent AI system, enabling supply chain planners to move from reactive firefighting to proactive decision-making — all from a single bilingual (English/Arabic) conversational interface.

At the heart of the platform is **Rashid** — an autonomous AI orchestrator that routes complex supply chain queries to specialist agents running in parallel. An S&OP planner can ask "What happens if our wheat supplier is delayed 21 days during Ramadan?" and Rashid will simultaneously dispatch five LLM agents to model demand impact, simulate inventory depletion, evaluate alternative suppliers, assess production capacity, and synthesize a mitigation plan — all in under 30 seconds.

Every page in the platform — from the KPI Dashboard to Scenario Planner to Report Builder — is a Rashid-powered workflow. The UI visualizes agent activity in real time, making the AI's reasoning transparent and auditable for board-level decisions.

Tagline: "In pursuit of better" | "بَحْثًا عَنِ الْأَفْضَلِ"

2. The Problem

AGI Food Division's supply chain spans commodity procurement (wheat, oils, rice, sugar from 8+ global origins), high-volume milling operations, multi-channel distribution (retail, foodservice, industrial), and regulatory compliance (ESMA, municipality standards). The S&OP team faces acute challenges:

2.1 Core Challenges

Challenge	Impact
Reactive planning — Stockouts and excess inventory detected after the fact	Fill rate drops below 97% target; AED 500K+ monthly in lost sales
Fragmented data — Demand, inventory, supplier, and production data in separate systems	No unified view; S&OP meetings rely on stale Excel snapshots
Manual scenario analysis — What-if scenarios require days of spreadsheet modeling	Decisions made without quantitative risk assessment
Report generation overhead — Weekly S&OP decks, inventory reviews, and scorecards consume 6+ hours	Senior planners spend 40% of time on formatting, not analysis
Limited visibility — No real-time view of cascading supply chain risks	Supplier delays discovered when production lines idle
Language barriers — Arabic-speaking leadership reviews English-only reports	Communication friction between planners and C-suite

2.2 The AGI Food Complexity

Unlike a single-product company, AGI Food must simultaneously manage:

Dimension	Scale
Active SKUs	150+ across 7 categories
Production Lines	12 lines across 3 plants
Supplier Network	8 global suppliers (Ukraine, USA, Brazil, Singapore, France, Canada, Malaysia, China)
Lead Time Range	10–45 days depending on origin
Demand Variability	30–50% seasonal swings (Ramadan, summer)
Service Level Target	>97% fill rate with <2% stockout

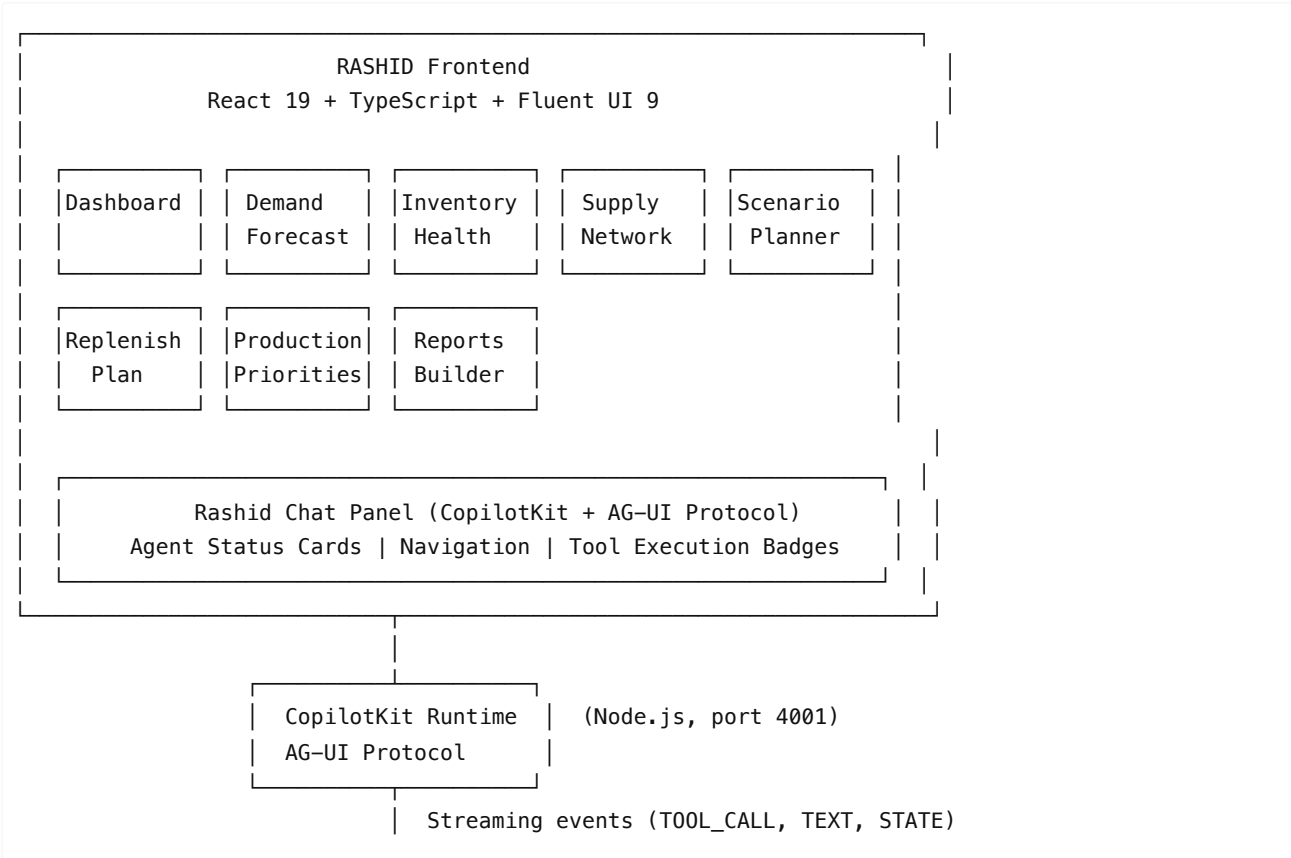
No existing supply chain tool combines real-time KPI monitoring, conversational AI, multi-agent scenario simulation, and automated report generation in a single platform. RASHID solves this.

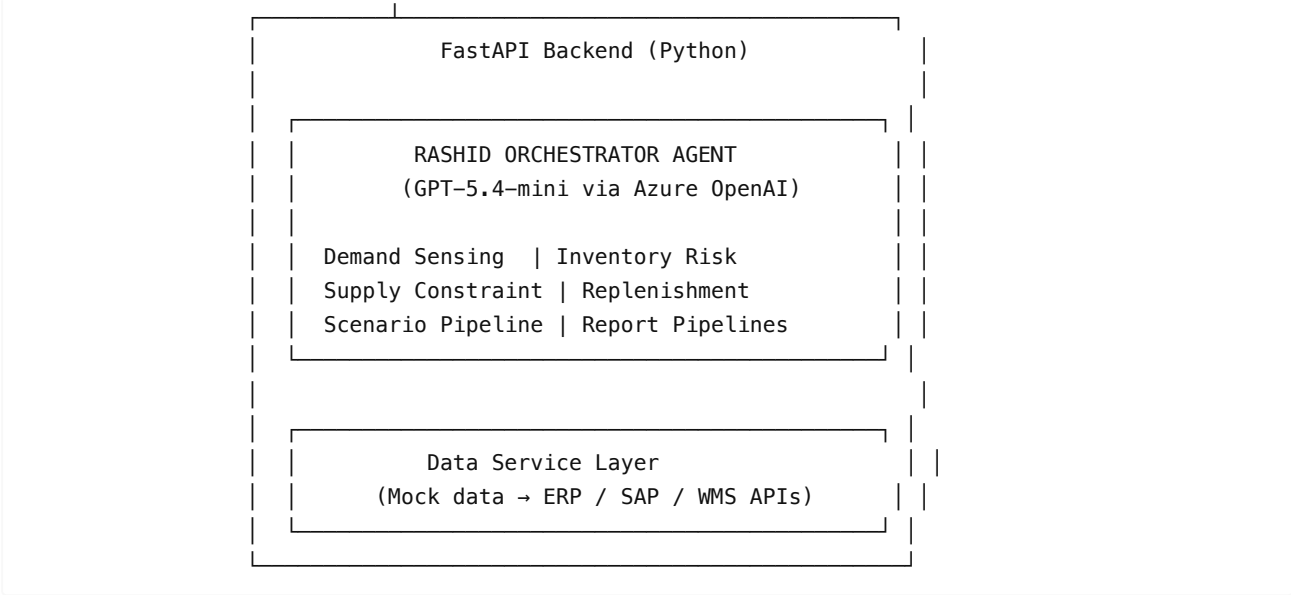
3. Solution Architecture

3.1 Design Philosophy

RASHID is built on the principle that **AI should drive the application, not decorate it**. Rashid is not a chatbot sidebar — it is the central nervous system that powers every workflow. Each page in the platform is a visualization layer over Rashid's agent outputs. When a user runs a Scenario Analysis, Rashid's 5-stage pipeline does the work. When the Report Builder produces an S&OP deck, Rashid's multi-agent pipeline built it. When Inventory Health flags critical items, Rashid's risk engine scored them.

3.2 Platform Architecture





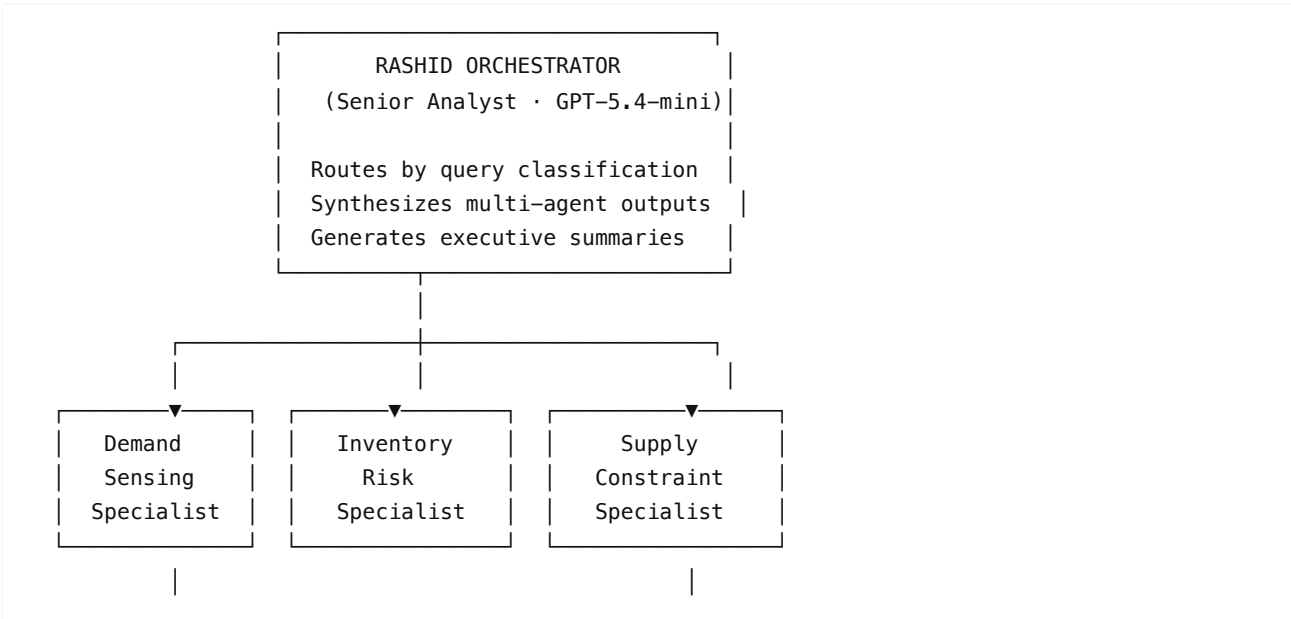
4. Multi-Agent Architecture — Rashid Orchestrator

4.1 Agent Identity

Property	Value
Name	Rashid (رشيد)
Meaning	"The Wise/Prudent" in Arabic
Role	Senior Supply Chain Analyst & Orchestrator
Model	GPT-5.4-mini via Azure OpenAI
Protocol	AG-UI (Agent-to-UI) for real-time streaming
Personality	Data-driven, executive-concise, action-oriented

4.2 Specialist Agents

Rashid delegates work to four specialist agents that run in parallel:





Agent	Responsibility	Key Outputs
Demand Sensing	POS sell-out, order history, seasonality, promotional lift	SKU-level 8-week forecasts with confidence bands
Inventory Risk	Stock positions, days-of-supply, risk scoring, ABC/XYZ classification	Risk matrix (critical/warning/normal/excess), shelf-life tracking
Supply Constraint	Supplier lead times, production capacity, logistics bottlenecks	Reliability scores, delayed PO flags, capacity utilization
Replenishment	Purchase orders, production priorities, safety stock adjustments	Prioritized action cards with confidence scoring & KPI impact

4.3 Routing Intelligence

Rashid classifies every user query and routes to the appropriate specialist:

Query Type	Route
Demand / forecast question	Demand Sensing → Suggest Actions
Stock / inventory status / risk	Inventory Risk → Suggest Actions
Supplier / capacity / logistics	Supply Constraint → Suggest Actions
What to order / replenishment	Replenishment → Suggest Actions
Full S&OP review	All 4 agents in parallel
What-if / scenario	Scenario Analysis Pipeline (5-stage)
Generate deck / report	Report Generation Pipeline (multi-agent)
SKU / Supplier / Plant detail	Entity Detail Lookup → Detail Drawer

5. Feature Deep-Dive — Nine Modules

5.1 KPI Dashboard

Real-time operational health across all supply chain dimensions:

- **6 KPI Cards** — Forecast Accuracy (MAPE), Inventory DOS, Fill Rate, Stockout Rate, Critical Alerts, Production Utilization
- **Risk Matrix** — Pie chart: critical, warning, normal, excess inventory distribution
- **Critical Alerts** — Top 4 active alerts with severity badges and affected SKUs
- **Quick Actions** — One-click: Morning Brief, Check Risks, Generate Report

KPI Targets:

KPI	Target	Status Logic
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Forecast Accuracy (MAPE)	< 15%	Green $\leq$ 15%, Red > 15%
Inventory Days of Supply	14–21 days	Green 14-21, Red outside
Fill Rate	> 97%	Green $\geq$ 97%, Red < 97%
Stockout Rate	< 2%	Green $\leq$ 2%, Red > 2%
Obsolescence Rate	< 1.5%	Monitoring
Production Utilization	70–85%	Optimal range

5.2 Demand Forecast

AI-powered demand intelligence with confidence visualization:

- **SKU Selection** — Filter by product, category, or channel
- **8-Week Forecast Chart** — Point forecast with 80% and 95% confidence bands (area chart)
- **Forecast Statistics** — High/medium confidence periods, average weekly demand
- **Demand Signals** — Trend direction, seasonality patterns, active promotions
- **Historical Context** — 12-week demand history for pattern validation

AGI Context: During Ramadan peak, the flour category sees 30-40% demand uplift. RASHID's confidence bands widen pre-Ramadan, signaling the planner to build safety stock.

5.3 Inventory Health

Portfolio-level inventory risk assessment:

- **Risk Matrix Visualization** — Distribution across critical (stockout imminent), warning (attention needed), normal (healthy), excess (overstock)
- **Position Table** — SKU, category, DOS, stock level, risk badge, shelf-life remaining
- **DOS-Sorted View** — Lowest days-of-supply first to highlight at-risk items
- **Drill-Down** — Click any SKU to open the Detail Drawer with full inventory profile

AGI Context: Cooking oil SKUs from Malaysian suppliers (14-day lead time) require different safety stock levels than Ukrainian wheat (28-40 day lead time). RASHID adjusts risk scoring per lead-time band.

5.4 Supply Network

Supplier performance monitoring and production capacity visualization:

- **Supplier Table** — Name, country, lead time (days), reliability score, active orders
- **Production Capacity** — Horizontal bar chart of line utilization with color-coded thresholds (red >85%, orange >70%)
- **Detail Panels** — Click supplier for performance trends, certifications, PO history; click line for maintenance schedule, production runs

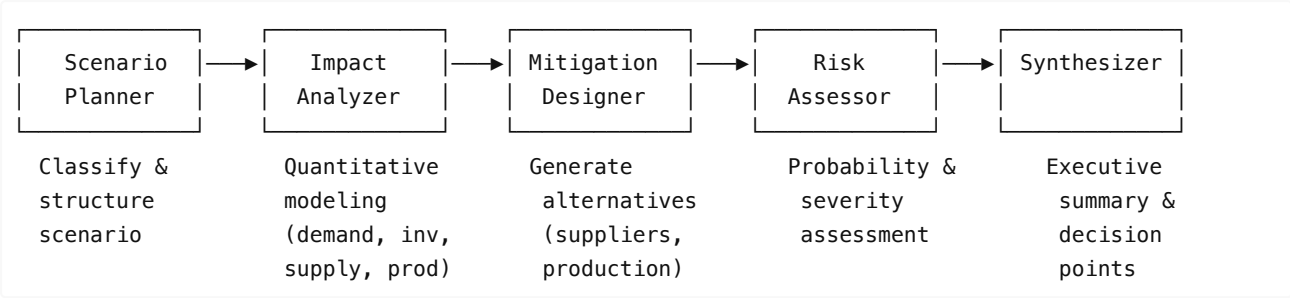
AGI Context: RASHID tracks 8 global suppliers across Ukraine, USA, Brazil, Singapore, France, Canada, Malaysia, and China — with reliability ranging from 72% (Black Sea Grain) to 95% (Cargill).

5.5 Scenario Planner (What-If Analysis)

The flagship analytical capability — a 5-stage LLM pipeline for quantitative scenario simulation:

- **Quick Scenario Chips** — Pre-built common scenarios (demand spike, supplier delay, production disruption, promotion, multi-factor)
- **Custom Natural Language Input** — "What if wheat prices spike 25% and our main flour supplier is delayed 3 weeks during Ramadan?"
- **Real-Time Pipeline Progress** — Visual 5-stage tracker with status badges

5-Stage Pipeline:



Result Tabs:

Tab	Content
Summary	Affected SKU count, KPI comparison (baseline vs. projected), risk narrative, recommended actions
Affected SKUs	Severity-coded table with demand delta %, weeks-to-stockout, DOS projection
Timeline	8-week demand trajectory, inventory depletion curves, stockout risk visualization
Mitigation	Prioritized options with cost (AED), fill-rate recovery, lead time; alternative suppliers; production surge capacity

AGI Context: Before Ramadan 2026, the S&OP team simulates "40% demand spike in flour + 21-day supplier delay" to quantify stockout risk and pre-approve mitigation actions — replacing 3 days of manual modeling with a 30-second AI pipeline.

5.6 Replenishment Plan

AI-generated, action-ready replenishment recommendations:

- **Prioritized Action Cards** — Each card shows SKU, urgency badge (critical/high/normal), recommended quantity, confidence level
- **Action Types** — Purchase order, production priority, safety stock adjustment, expedite
- **Scenario Variants** — Conservative, balanced, aggressive options with KPI trade-offs
- **Approve/Dismiss Workflow** — One-click approval with KPI impact confirmation dialog
- **Rationale Text** — AI-generated explanation for each recommendation

AGI Context: "Order 200 MT of wheat flour from Cargill (95% reliable, 21-day lead). Rationale: DOS drops below 10 days by Week 3 if current consumption continues. KPI impact: +2.1% fill rate recovery."

5.7 Production Priorities

Real-time production line monitoring and scheduling:

- **Utilization Chart** — All 12 lines visualized with current utilization %
- **Schedule Table** — Line name, plant, capacity (MT/day), current SKU, shift pattern, planned maintenance
- **Capacity Headroom** — Spare capacity available for surge production
- **Detail Drill-Down** — Click any line for maintenance history, production runs, yield %

5.8 Report Builder (Multi-Agent Document Generation)

Fully agentic report generation with zero manual formatting:

Template	Format	Use Case
Weekly S&OP Review	PPTX	KPIs, alerts, demand outlook, actions (10 slides)

Inventory Status	PPTX	Stock positions, risk matrix, aging analysis
Demand Accuracy	PPTX	MAPE trends, bias analysis, forecast vs actuals
Executive S&OP Summary	DOCX/PDF	High-level performance & decisions for leadership
Inventory Deep-Dive	DOCX/PDF	Portfolio health, stockout risk, safety stock optimization
Replenishment Plan	XLSX	Priority orders, production schedule, cost analysis
Supplier Scorecard	XLSX	Lead times, reliability rankings, quality scores

Generation Pipeline (PPTX — 5 Agents):

Planner → Content Writer → Designer → Critic → Repair (up to 2 iterations)

Features:

- Section customization (toggle specific sections on/off)
- Audience configuration (S&OP Committee, Executive Leadership, Operations Team)
- Real-time progress visualization per pipeline stage
- Preview before download (slide thumbnails, section outlines, data tables)
- Regeneration with per-section feedback
- AGI branding with navy/magenta corporate identity

5.9 Chat Interface — Conversational Supply Chain

The Rashid chat panel is not a feature bolted on — it is the primary interaction mode:

- **Natural Language Queries** — "What's the stockout risk for cooking oil this month?"
- **Tool Execution Badges** — Real-time visual indicators showing which specialist is working
- **Automatic Navigation** — Asking about inventory automatically shows the Inventory Health page
- **Entity Detail Drawers** — "Tell me about supplier Cargill" opens the supplier panel without page navigation
- **Action Suggestions** — Post-response follow-up chips (Review, Approve, Generate Report)
- **Context Persistence** — Session-based conversation memory

Supported Natural Language Patterns:

User Intent	Rashid Action
"Morning brief"	Compile daily KPIs, critical alerts, focus areas
"Check supply alerts"	Surface stockout warnings, delivery delays
"Run scenario: 30% demand spike in oils"	Execute 5-stage scenario pipeline
"Generate weekly S&OP deck"	Launch 5-agent PPTX pipeline
"Tell me about SKU-001"	Open SKU detail drawer
"Show supplier Bunge MENA"	Open supplier detail drawer
"Compare conservative vs aggressive replenishment"	Side-by-side KPI comparison
"What should I focus on today?"	AI-prioritized recommendations

6. User Journeys

6.1 Morning Routine — S&OP Planner

1. Open RASHID → Dashboard shows 3 critical alerts, fill rate at 96.2% (below target)
2. Ask: "Morning brief" → Rashid compiles daily KPIs, flags 4 SKUs below safety stock
3. Click critical SKU → Detail Drawer shows 6 days DOS, declining trend
4. Ask: "What happens if this supplier is delayed 2 more weeks?"
→ Scenario pipeline runs: 14 SKUs affected, 3 will stock out by Week 4
5. Review mitigation tab → Approve expedite order from Cargill (21-day lead)
6. Ask: "Generate the weekly S&OP deck for tomorrow's meeting"
→ 10-slide PPTX generated with current data, downloaded in 45 seconds

6.2 Risk Escalation — Supply Chain Director

1. Receive Rashid alert: "Black Sea Grain delivery delayed 14 days (port congestion)"
2. Open Supply Network → See supplier reliability dropped to 72%
3. Ask: "Run scenario: 21-day delay from Black Sea Grain during Ramadan peak"
→ Pipeline shows 33 SKUs affected, AED 1.2M potential lost sales
4. Review alternative suppliers → Cargill has 50,000 MT spare capacity
5. Ask: "Generate executive summary of this scenario for the CFO"
→ DOCX generated with impact analysis, mitigation costs, recommendation
6. Approve the recommended split order: 60% Cargill, 40% Louis Dreyfus

6.3 Weekly S&OP Meeting — Cross-Functional Team

1. Generate pre-meeting deck (Weekly S&OP Review template)
2. During meeting, ask live questions: "Which categories have forecast accuracy below 20%?"
3. Deep dive: "Show demand forecast for pasta category" → Chart with confidence bands
4. Decision: "Run scenario: promote dairy +50% for 4 weeks" → Instant impact assessment
5. Post-meeting: "Generate replenishment plan Excel with today's decisions"
→ XLSX with priority orders, quantities, and timeline

6.4 New Supplier Evaluation — Procurement Manager

1. Ask: "Show me all suppliers with reliability below 80%"
→ Inventory Risk agent surfaces at-risk supplier dependencies
2. Click supplier → Detail drawer shows 6-month performance trend declining
3. Ask: "What's the impact if we lose this supplier entirely?"
→ Scenario: production_disruption, 45 SKUs affected
4. Review alternative suppliers with capacity → Identify 2 replacements
5. Generate supplier scorecard Excel for procurement committee

7. Technical Architecture

7.1 Technology Stack

Layer	Technology	Purpose
Frontend	React 19, TypeScript, Vite	SPA with HMR
UI Framework	Fluent UI 9	Microsoft design system (enterprise grade)
Charts	Recharts	Data visualization (area, line, bar, pie)
AI Chat	CopilotKit + AG-UI Protocol	Agent-UI streaming integration
Backend	FastAPI (Python 3.13)	API server with async support

AI Orchestration	Agent Framework + Azure OpenAI	Multi-agent routing and tool dispatch
LLM Model	GPT-5.4-mini (Azure)	All agent reasoning and generation
Document Gen	PptxGenJS, docx, jsPDF, ExcelJS	Client-side file rendering
State Sync	Custom event system + polling	Cross-component communication
Session	UUID-based session management	Multi-user isolation
i18n	Custom React context	English/Arabic with RTL support

7.2 Agent Framework Integration

RASHID uses the **AG-UI (Agent-to-UI) Protocol** for real-time agent communication:

- **Streaming Events** — Tool calls, text generation, and state updates stream to the frontend in real time
- **Tool Registration** — 15+ tools registered with typed parameters
- **State Updates** — Agents push structured results (scenarios, reports, forecasts) to frontend via polling endpoint
- **Session Isolation** — Each user session maintains independent agent state

7.3 Data Architecture

Data Source	Current	Production Target
SKU Master	Mock (150+ SKUs)	SAP MM
Inventory Positions	Mock (real-time)	WMS API
Demand Forecasts	Mock (8-week)	SAP APO / IBP
Supplier Data	Mock (8 suppliers)	SAP SRM / Ariba
Production Schedule	Mock (12 lines)	SAP PP / MES
Purchase Orders	Mock	SAP MM
Quality Results	Mock	LIMS
Alerts	Mock (rule-based)	Event stream / IoT

8. Bilingual Support (English / Arabic)

RASHID supports full bilingual operation:

Aspect	Implementation
UI Labels & Navigation	Translated via i18n context
Agent Responses	Language-aware prompting (responds in user's language)
RTL Layout	CSS direction switching for Arabic
Report Generation	LLM generates content in selected language
Chat Interface	Arabic input and response supported
Settings Toggle	One-click language switch

9. Expected Outcomes

9.1 Efficiency Gains

Metric	Before RASHID	With RASHID	Improvement
Scenario analysis time	2–3 days (manual Excel)	30 seconds	99% reduction
Weekly S&OP deck preparation	6+ hours	45 seconds	99% reduction
Morning supply review	45 minutes (multiple systems)	2 minutes (single query)	95% reduction
Stockout detection	After the fact	Predictive (8-week horizon)	Proactive vs. reactive
Supplier risk assessment	Monthly review cycles	Real-time monitoring	Continuous vs. periodic
Report formatting effort	40% of planner time	Zero (fully automated)	100% elimination

9.2 Strategic Impact

- **Decision Speed** — From weekly S&OP cycles to real-time scenario-based decisions
- **Risk Visibility** — Cascading risks surfaced before they materialize
- **Service Level** — Fill rate maintained >97% through predictive replenishment
- **Cost Optimization** — Safety stock right-sized per lead-time band (avoid over-stocking)
- **Stakeholder Communication** — Board-ready documents generated on demand in either language

10. Roadmap Considerations

Phase	Scope	Timeline
Phase 1 (Current)	Prototype with mock data, full UI, all agent pipelines	Complete
Phase 2	SAP/ERP data integration, live inventory feeds	Q3 2026
Phase 3	IoT/sensor integration (warehouse temperature, line sensors)	Q4 2026
Phase 4	Automated PO creation (agent-initiated procurement)	Q1 2027
Phase 5	Cross-division intelligence (link to BASIRAH market intel)	Q2 2027

11. Security & Classification

Aspect	Current (Prototype)	Production Recommendation
Authentication	Session-based login	Azure AD / SSO integration
Data Classification	Internal Confidential	Per-document classification markings
API Security	Session headers	OAuth 2.0 + API key rotation
LLM Data	Azure OpenAI (data stays in Azure tenant)	Private endpoint, no external data leakage
Audit Trail	Session-based	Full action audit log with user attribution

12. Conclusion

RASHID transforms AGI Food Division's supply chain planning from a fragmented, reactive, spreadsheet-driven process into a unified, AI-powered decision engine. By combining:

- **Multi-agent AI orchestration** — 4 specialist agents + orchestrator for parallel intelligence gathering
- **Quantitative scenario simulation** — 5-stage LLM pipeline replacing days of manual modeling
- **Automated document generation** — 7 templates across PPTX/DOCX/XLSX with zero formatting effort
- **Conversational interface** — Natural language access to all supply chain data and actions
- **Real-time risk monitoring** — Predictive alerts before stockouts materialize

...RASHID enables AGI's S&OP team to make faster, better-informed decisions — protecting a >97% fill rate across 150+ SKUs while managing a global supplier network spanning 8 countries with lead times from 10 to 45 days.

The platform is designed for progressive deployment: immediate value from the prototype with mock data, scaling to full ERP integration without architectural changes.

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